

Gaussian product blocking: the tree-level coupling never moves

Result, 2026-09-28 (Claude; written derivation, to be refereed). For the free (Gaussian) lattice gauge field in D dimensions with heat time t , one isotropic product-blocking step $a \mapsto 2a$ maps the law of the fluxes to a Gaussian law whose covariance at small coarse momentum K is

$$C'(K) = 2^{4-D} t [P_{\text{cont}}(\hat{K}) + O(K)]$$

on exact 2-forms, with $O(K^2)$ in plaquette-centred phase conventions (Theorem G1): the zero-momentum coefficient is exactly the heat time $\lambda_D(2a)^{4-D}$ of the coarse lattice. The statement holds for any input covariance supported on exact 2-forms with the same small-momentum form, so it iterates: along any number of steps, and for the four directional halvings of one step in any order (Fubini), the tree-level coupling of the Gaussian part is preserved exactly. In 1 + 3 it does not run at all.

Consequence for cell 4 (the four-dimensional note, (15) and the Remark after its telescoping paragraph). The endpoint term $c_{\text{in}} - c_{\text{out}}$ receives nothing from the zero-momentum part of the Gaussian kernel. It arises only from the dependence of the one-loop determinant on the kernel's shape (its $O(K^2)$ and higher terms) and from the generated cubic and quartic vertices. Along an iteration the shape may drift; the average rate $-2b_0 \log 2$ per step needs only $c(\zeta_n, \mathcal{K}_n) = o(n)$, so even a logarithmic drift of shape parameters would be harmless.

Caution. Convergence of the full kernel under iterated sharp product blocking is not claimed. Product blocking makes coarse fluxes the fluxes of the fine field through sharp surfaces, whose variances in the continuum carry boundary (perimeter-type) divergences; the perfect-action literature smears the blocking to obtain a local fixed point (Bell and Wilson 1975; for gauge fields Bietenholz and Wiese 1996; metadata).

1. Setting

Take $G = \mathbb{R}$ (colour components decouple in the Gaussian theory) on a periodic lattice of spacing a , link variables A , fluxes $\phi = dA$ and weight $\prod_p e^{-\phi_p^2/(2t)}$. Modulo gauge, ϕ ranges over the exact 2-forms, with the flat Gaussian density there, so its covariance is tP , where $P(k)$ is the orthogonal projector onto exact 2-forms at lattice momentum k ($k \neq 0$; the harmonic sector at $k = 0$ is set aside). Product blocking sets the coarse link to the sum of the two fine links along it; by Stokes the coarse flux through a coarse plaquette in the plane $(\mu\nu)$ is the sum of the four fine fluxes inside it, $(B\phi)_{\mu\nu}(X) = \sum_{a,b \in \{0,1\}} \phi_{\mu\nu}(2X + ae_\mu + be_\nu)$. This is the Gaussian case of the pushforward of Proposition 1 of the series/parallel note, composed over the D directions.

2. Theorem G1 and its proof

The alias sum. With unitary Fourier transforms on the fine and coarse tori, the coarse momenta $K \in (-\pi, \pi]^D$ (units $1/2a$) receive the 2^D fine momenta $k_n = K/2 + \pi n$, $n \in \{0, 1\}^D$, and

$$(B\phi)_{\mu\nu}^\wedge(K) = 2^{-D/2} \sum_n f_{\mu\nu}(k_n) \hat{\phi}_{\mu\nu}(k_n), \quad f_{\mu\nu}(k) = (1 + e^{ik_\mu})(1 + e^{ik_\nu}).$$

Hence the coarse covariance is $C'(K)_{\mu\nu, \rho\sigma} = 2^{-D} t \sum_n f_{\mu\nu}(k_n) \overline{f_{\rho\sigma}(k_n)} P(k_n)_{\mu\nu, \rho\sigma}$.

The term $n = 0$. $|f_{\mu\nu}(K/2)|^2 = 16 \cos^2(K_\mu/4) \cos^2(K_\nu/4) = 16(1 + O(K^2))$, and the lattice projector satisfies $P(k) = P_{\text{cont}}(\hat{k}) + O(k)$, homogeneous of degree zero in the direction $\hat{k}$; the $O(k)$ is a diagonal phase, and with plaquette-centred conventions (symbols $2 \sin(k_j/2)$) the error is $O(k^2)$. So the $n = 0$ term is $2^{4-D} t [P_{\text{cont}}(\hat{K}) + O(K)]$.

The terms $n \neq 0$. If $n_\mu = 1$ or $n_\nu = 1$, then $1 + e^{i(\pi + K_\mu/2)} = 1 - e^{iK_\mu/2} = O(K)$, so $f_{\mu\nu}(k_n) = O(K)$: a term whose two form factors are both $O(K)$ is $O(K^2)$, and a mixed term with an $O(1)$ partner is covered by the closedness bound below. If $n_\mu = n_\nu = 0$ but $n_\lambda = 1$ for some $\lambda \notin \{\mu, \nu\}$, use closedness: exact forms are closed, and the $(\mu\nu\lambda)$ component

of $d\phi$ at k_n reads $d_\mu \hat{\phi}_{\nu\lambda} - d_\nu \hat{\phi}_{\mu\lambda} + d_\lambda \hat{\phi}_{\mu\nu} = 0$ with $d_j = e^{ik_j} - 1$, where $d_\mu, d_\nu = O(K)$ and $d_\lambda = -2 + O(K)$. So on the range of $P(k_n)$ the component $\hat{\phi}_{\mu\nu}$ is $O(K)$ times the others. Since P is an orthogonal projector, $\|Pe_{\mu\nu}\|^2 = (Pe_{\mu\nu})_{\mu\nu} = O(K)\|Pe_{\mu\nu}\|$, so $\|P(k_n)e_{\mu\nu}\| = O(K)$. An entry $P_{\mu\nu,\rho\sigma} = \langle Pe_{\mu\nu}, Pe_{\rho\sigma} \rangle$ is then $O(K)$ times $\|Pe_{\rho\sigma}\|$, which is itself $O(K)$ when $n_\rho = n_\sigma = 0$ (the same λ serves), while otherwise $f_{\rho\sigma}(k_n) = O(K)$ supplies the second factor. Every term with $n \neq 0$ is $O(K^2)$. $\square$

Iteration. The proof used only that the input covariance is supported on exact 2-forms (for closedness at k_n) and has the form $t[P_{\text{cont}} + O(k)]$ at small k (for the $n = 0$ term). The output has both properties, so by induction every step multiplies the zero-momentum heat time by exactly 2^{4-D} . The four directional Gaussian integrations of one step compose, in any order, to the isotropic one (Fubini), so the exact directional defects of Proposition 2 and Corollary 2_s compose to a kernel with the same zero-momentum coefficient; this is the free-field statement that their defects vanish at zero momentum and constant flux.

3. What the theorem leaves to the interacting theory

In the abelian theory the coupling has no loop corrections, and G1 is the whole story at zero momentum. In the non-abelian theory the one-loop matching coefficient $c(\zeta, \mathcal{K})$ of (15) depends on the full kernel, and G1 localizes the obstruction: the finite endpoint term comes from how the loop integrals feel the shape of the generated quadratic kernel and from the generated vertices, never from a drift of the tree-level coupling. Whether the shape parameters stay bounded, or drift slowly, under the iteration is the question that the Remark (many steps) of the four-dimensional note needs answered only in the weak form $o(n)$.

4. Consequence for STATE

Atlas cell 4: the Gaussian half of the composition obstruction is closed at zero momentum (Theorem G1, iterated). The open half is the one-loop dependence on the kernel shape and on generated vertices, with the average rate requiring only $c = o(n)$ along the iteration.